

MADISON COOTS

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RESEARCH FOCUS

Responsible AI, inequality, discrimination, and the use of computational methods to advance decision-making and policy across a variety of domains, including in healthcare, lending, criminal justice, and education.

EDUCATION

Harvard Kennedy School <i>Ph.D. in Public Policy</i>	Cambridge, MA August 2022 - Present
Harvard University <i>M.A. in Public Policy</i>	Cambridge, MA August 2022 - May 2025
Stanford University <i>M.S. in Computer Science</i> <i>Specialization: Artificial Intelligence</i>	Stanford, CA September 2019 - June 2021
Stanford University <i>Bachelor of Science - Management Science and Engineering</i> <i>Minor: English</i>	Stanford, CA September 2015 - June 2019

PEER-REVIEWED PUBLICATIONS

- [1] **Automated Court Date Reminders Reduce Warrants for Arrest: Evidence from a Text Messaging Experiment.** Alex Chohlas-Wood, **Madison Coots**, Joe Nudell, Julian Nyarko, Emma Brunskill, Todd Rogers, and Sharad Goel. *Science Advances*. 2025.
- [2] **Racial Bias in Clinical and Population Health Algorithms: A Critical Review of Current Debates.** **Madison Coots**, Kristin A. Linn, Sharad Goel, Amol S. Navathe, and Ravi B. Parikh. *Annual Review of Public Health*. 2025.
- [3] **A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk.** **Madison Coots**, Soroush Saghaian, David Kent, and Sharad Goel. *Annals of Internal Medicine*. 2024.
- [4] **Learning to be Fair: A Consequentialist Approach to Equitable Decision-Making.** Alex Chohlas-Wood, **Madison Coots**, Henry Zhu, Sharad Goel, and Emma Brunskill. *Management Science*. 2024.
- [5] **Designing Equitable Algorithms.** Alex Chohlas-Wood*, **Madison Coots***, Julian Nyarko*, and Sharad Goel*. *Nature Computational Science*, Vol. 3. 2023. *Authors contributed equally.

RESEARCH IN PROGRESS

- [1] **Algorithmic Bias in Lending: Evidence from a Fintech Audit.** **Madison Coots**, Robert Bartlett, Julian Nyarko, and Sharad Goel. *Working paper*. 2026.
- [2] **Incorporating Text in Empirical Audits of Police Searches.** **Madison Coots**, Hans Gaebler, Josh Grossman, Alex Chohlas-Wood, Julian Nyarko, and Sharad Goel.
- [3] **AI Tutoring and Teacher Support Tools for Elementary Education**, with Calvin Isley, Elisa Xi Chen, Nick Masi, Joe Nudell, Charlotte Tuminelli, Teddy Svoronos, and Sharad Goel.

HONORS, AWARDS, AND FELLOWSHIPS

Dissertation Research Grant, \$10,000 Horowitz Foundation for Social Policy	2026
Rising Star in IOE-ISyE-MS&E Workshop jointly hosted by University of Michigan, Georgia Tech, and Stanford University	2026
Dean's Award for Excellence in Student Teaching Harvard Kennedy School of Government	2025

Distinction in Student Teaching Award	2025
• Harvard Kennedy School of Government	
Social Equity and Health Equity Grant, \$5,000	2025
• Malcolm Weiner Center for Social Policy, Harvard Kennedy School of Government	
James M. and Cathleen D. Stone PhD Scholar in Inequality and Wealth Concentration	2023 - Present
• Stone Program in Wealth Distribution, Inequality, and Social Policy, Harvard University	
Harvard Graduate Prize Fellowship	2022 - 2023
• Harvard University	
Stanford Engineering Coterminial Fellowship	2019 - 2020
• Stanford University	
Stokes Scholar	2017 - 2021
• U.S. Federal Government	

SELECTED PRESENTATIONS

Production and Operations Management Society Annual Conference	Reno, NV
“A Profit-Based Measure of Lending Discrimination” (Submitted talk)	2026
IOE-ISyE-MS&E Rising Stars Workshop	Ann Arbor, MI
“A Profit-Based Measure of Lending Discrimination” (Invited talk and poster)	2026
American Law and Economics Association Conference	New York, NY
“A Profit-Based Measure of Lending Discrimination” (Submitted talk)	2025
Production and Operations Management Society Annual Conference	Atlanta, GA
“A Profit-Based Measure of Lending Discrimination” (Invited talk)	2025
Predictive Analytics & Comparative Effectiveness Center Symposium	Boston, MA
“A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk” (Invited talk)	2024
Society of Medical Decision Making 46th Annual Meeting	Boston, MA
“A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk” (Submitted talk)	2024
International Conference on Computational Social Science	Philadelphia, PA
“A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk” (Poster)	2024
Computational and Methodological Statistics Conference	Remote
“A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk” (Invited talk)	2023
APPAM Fall Research Conference	Atlanta, GA
“Automated Court Date Reminders Reduce Warrants for Arrest” (Submitted talk)	2023
INFORMS General Meeting	Phoenix, AZ
“A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk” (Invited talk)	2023
INFORMS Healthcare Conference	Toronto, ON
“A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk” (Invited talk)	2023
ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization	Arlington, VA
“Learning to be Fair” (Submitted talk with Alex Chohlas-Wood)	2022

TEACHING

Harvard Kennedy School of Government
• Teaching Fellow. API 201: Quantitative Analysis and Empirical Methods I (Fall 2024, 2025). <i>Graduate course in applied statistics: exploring and summarizing data with R, probability theory, statistical inference, decision analysis.</i>
• Teaching Fellow. API 202Z: Quantitative Analysis and Empirical Methods II (Spring 2025, 2026). <i>Advanced graduate course in applied statistics: statistical inference and linear regression models.</i>
• Teaching Fellow. API 203Z: Quantitative Analysis and Empirical Methods III (Spring 2025). <i>Advanced graduate course in applied statistics: causal inference and machine learning.</i>
• Teaching Fellow. DPI 681M: The Science and Implications of Generative AI (Spring 2024) <i>Designed and taught the technical complement to the course on language models, deep learning models, and transformers.</i>

Stanford University, School of Engineering

- **Course Assistant.** MS&E 125: Applied Statistics (Winter 2020) *Undergraduate course in applied statistics: exploring and summarizing data, statistical inference, linear and logistic regression models.*
- **Course Assistant.** MS&E 252: Foundations of Decision Analysis (Fall 2019). *Graduate course in quantitative decision analysis covering: utility theory, sensitivity analysis, value of information. Recognized by Stanford Center for Professional Development for excellence in teaching.*

Stanford Code in Place Program

- **Section Leader.** (Spring 2020). *Part of a teaching team for Code in Place, offered by Stanford during COVID-19 pandemic, with 10,000 global students and 900 volunteer teachers participating from around the world. Prepared and taught a weekly discussion section of 10-12 students to supplement professors’ lectures in a 5-week introductory online Python programming course.*

PROFESSIONAL SERVICE

Peer Review

Ad Hoc Reviewer

- *Epidemiology* (2025)
- *International Conference on Computational Social Science* (2025, 2026)

SKILLS SUMMARY

Languages: R, Python

AI: LLM prompt and response evaluation, text data processing and information extraction, chatbot design and user experience for AI systems

Methods: Stochastic modeling, decision analysis, statistical inference, causal inference

PROFESSIONAL EXPERIENCE

Senior Data Scientist (Part-time) <i>Systems & Technology Research</i>	Woburn, MA <i>Feb. 2023 - Present</i>
Data Scientist <i>Stanford Computational Policy Lab</i>	Boston, MA <i>Sep. 2020 - Aug. 2022</i>
Data Scientist (Part-time) <i>Aerospace Technical Services</i>	Remote <i>Sep. 2020 - Jun. 2024</i>
Data Science Intern <i>U.S. Federal Government (agency withheld)</i>	Washington D.C. <i>Jun. 2017 - Jan. 2021</i>